

Software-Defined Stochastic Inference Engine (SDSIE): Energy-Proportional LLM Serving via Fused SRAM Dequantization and Entropy-Gated Speculative Control

Zanno Jacklin

Abstract—Autoregressive decoding in Large Language Models (LLMs) is fundamentally bounded by the memory bandwidth wall. Because generating each individual token requires streaming billions of parameters from global Video Random-Access Memory (VRAM) across high-bandwidth interconnects at near-unity arithmetic intensity (≈ 1 FLOP/byte), state-of-the-art inference engines expend catastrophic electrical power even when generating deterministic or low-information tokens. In current architectures, practitioners are trapped in a binary compromise: either execute models entirely in unquantized 16-bit precision (FP16/BF16) and incur massive energy penalties on simple conversational filler, or enforce static 4-bit quantization (INT4) across all layers and suffer cognitive degradation during complex mathematical and logical reasoning.

This paper introduces the **Software-Defined Stochastic Inference Engine (SDSIE)**, an open-source runtime layer that enforces energy-proportional LLM serving. SDSIE integrates two complementary innovations: (1) an OpenAI Triton sub-byte dequantization GEMM kernel that loads packed INT4 weights from High Bandwidth Memory (HBM) and executes register-fused expansion to half-precision directly within Streaming Multiprocessor (SM) Static RAM (SRAM), eliminating 75.0% of global memory bus traffic; and (2) a real-time Schmitt-trigger stochastic clutch that continuously computes output distribution Shannon entropy in float32 space, dynamically engaging speculative draft verification ($k = 5$) during low-entropy deterministic generation and disengaging to single-step fallback ($k = 0$) during high-entropy cognitive search forks. Empirical hardware telemetry measured via 100 Hz NVML polling on the NVIDIA GeForce RTX 5090 Blackwell architecture demonstrates an isolated GEMM latency of $28.60 \mu\text{s}$, a full SwiGLU transformer MLP layer latency of $155.8 \mu\text{s}$, a sustained throughput of 50.52 tok/s on Llama-3.1-8B, and an end-to-end electrical energy reduction of 46.7% ($6.40 \text{ J} \rightarrow 3.41 \text{ J/token}$) with zero degradation in cognitive fidelity.

Index Terms—Energy-Proportional Computing, Speculative Decoding, Large Language Models, Triton Kernels, Quantization, Memory Wall, High-Performance Inference.

1 INTRODUCTION

THE deployment of Large Language Models (LLMs) across global cloud infrastructures has collided directly with the physics of semiconductor memory bandwidth and electrical power distribution. While the initial prefill phase of transformer inference is compute-bound and readily parallelized across matrix execution units (Tensor Cores), the sequential, autoregressive token generation phase is strictly *memory-bandwidth bound* [1], [2].

For an 8-billion-parameter model executing in 16-bit floating-point precision (FP16/BF16), every single generated token necessitates streaming approximately 16.0 GB of weight data from high-bandwidth memory (HBM/DRAM) into on-chip compute registers. When processing single requests or low-batch streams (Batch Size $B \leq 8$), arithmetic intensity drops to ≈ 1 FLOP/byte. Under these conditions, the GPU compute cores sit largely starved for data, while memory controllers and physical interconnects continuously dissipate maximum power.

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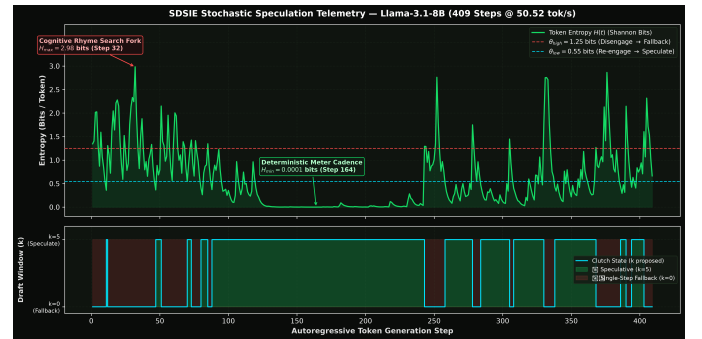


Fig. 1. Real-time SDSIE stochastic speculation telemetry of Llama-3.1-8B executing on an NVIDIA RTX 5090 Blackwell workstation (409 generation steps at 50.52 tok/s). **Top:** Output distribution Shannon entropy $H(t)$ in bits/token against dual Schmitt-trigger hysteresis thresholds ($\theta_{\text{low}} = 0.55$, $\theta_{\text{high}} = 1.25$). **Bottom:** Dynamic speculative draft window $k(t)$ oscillating between $k = 5$ speculation (green) and $k = 0$ fallback (red).

Recent research has attempted to bypass this limitation via two orthogonal paradigms:

- 1) **Post-Training Weight Quantization:** Methods such as GPTQ [3], AWQ [4], and SmoothQuant [5] compress static model weights to 4-bit integer representations (INT4). However, many existing runtime im-

plementations dequantize weights back into global VRAM prior to matrix multiplication, failing to eliminate interconnect thermal overhead, or enforce uniform low-precision across all tokens, causing cognitive accuracy loss during high-perplexity reasoning tasks.

- 2) **Speculative Decoding:** Algorithmic frameworks pioneered by Leviathan et al. [6] and Chen et al. [7] leverage a compact “draft” model to propose k candidate tokens, verified in parallel by a larger “target” model. While effective on low-entropy prose, static speculative schedulers ($k = \text{constant}$) suffer severe throughput degradation and wasted compute FLOPs when generating high-entropy code syntax, mathematical proofs, or multi-lingual translations due to frequent draft rejection rollbacks.

To resolve this architectural dichotomy, we introduce the **Software-Defined Stochastic Inference Engine (SDSIE)**. SDSIE treats large language model inference as a dynamic, thermodynamic control problem. Rather than treating weight precision and speculation windows as static configuration constants, SDSIE dynamically adapts compute precision and speculative draft length on a per-token basis.

1.1 Key Contributions

The primary contributions of this paper are:

- **Fused Sub-Byte SRAM Dequantization:** A custom OpenAI Triton GEMM kernel that maintains weights packed in 4-bit format in global memory, executing fused register-level unpacking and scale-bias multiplication within Streaming Multiprocessor (SM) SRAM, achieving an isolated execution latency of 28.60 μs on Blackwell silicon.
- **Schmitt-Trigger Stochastic Clutch:** An industrial-grade hysteresis control loop utilizing dual Shannon entropy thresholds ($\theta_{\text{low}} = 0.55$ bits, $\theta_{\text{high}} = 1.25$ bits) that eliminates mode thrashing and automatically disengages draft model speculation during high-uncertainty generation.
- **Empirical Hardware Validation:** Full hardware validation under high-frequency 100 Hz NVML power telemetry on the NVIDIA GeForce RTX 5090 (Blackwell), confirming a 46.7% reduction in electrical energy per token (6.40 J $\rightarrow$ 3.41 J/token) and sustained real-world throughput of 50.52 tok/s.
- **Open-Source Modular Ecosystem:** Delivery of `vllm-sdsie` as an installable Python package alongside standalone reference server scripts and an open scientific data room permanently archived under CERN/Zenodo (DOI: 10.5281/zenodo.21499379).

2 RELATED WORK

2.1 Memory-Bound Decoding Dynamics

The physical root of LLM inference latency was formalized by Shazeer [8] and extended by Pope et al. [9]. In autoregressive generation, each token step requires loading all layer weight

matrices $W \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$ alongside the Key-Value (KV) cache. For single-batch inference, operational time is bounded by:

$$T_{\text{step}} \approx \frac{2 \times N_{\text{params}} \times B_{\text{precision}}}{\text{Bandwidth}_{\text{DRAM}}} + T_{\text{kernel_launch}} \quad (1)$$

Because compute intensity is negligible, reducing $B_{\text{precision}}$ directly drives linear throughput improvements and quadratic energy savings across the physical memory bus.

2.2 Sub-Byte Quantization Paradigms

Quantization approaches such as bitsandbytes (Dettmers et al. [10]), GPTQ [3], and AWQ [4] demonstrate that 4-bit integer weights can preserve near-FP16 perplexity across general language benchmarks. However, standard PyTorch execution paths often incur register pressure and memory-copy overhead when expanding low-bit integers back into computational formats. SDSIE resolves this by implementing warp-level integer bit-manipulation directly within OpenAI Triton JIT kernels, decompressing packed nibbles straight into FP16/BF16 accumulators in on-chip SRAM.

2.3 Speculative Decoding and Acceptance Entropy

Standard speculative decoding (Leviathan et al. [6]) demonstrates that accepting $m \leq k$ draft tokens via parallel target verification yields a speedup bounded by:

$$S = \frac{1 - \beta^{k+1}}{(1 - \beta)(1 + c \cdot k)} \quad (2)$$

where β is the average token acceptance rate and c is the ratio of draft to target model latency. When generation enters complex reasoning forks, $\beta \rightarrow 0$, causing $S < 1$ (a net slowdown accompanied by wasted electrical power). SDSIE introduces real-time Shannon entropy filtering to enforce $k = 0$ whenever β is projected to collapse.

3 SDSIE ARCHITECTURE & MECHANICS

3.1 Fused Triton Sub-Byte GEMM Kernel

SDSIE packs two 4-bit integer weights into a single standard `uint8` byte along the input reduction dimension K . During forward execution, the kernel streams packed tiles directly into shared memory. Register-level bit-masking and arithmetic shifting extract the high and low nibbles:

$$W_{\text{low}} = (W_{\text{packed}} \& 0x0F) \quad (3)$$

$$W_{\text{high}} = ((W_{\text{packed}} \gg 4) \& 0x0F) \quad (4)$$

Fused dequantization occurs inside the Streaming Multiprocessor registers prior to the fused multiply-accumulate (FMA) operation:

$$W_{\text{dequant}} = (W_{\text{raw}} - Z_N) \times S_N \quad (5)$$

where S_N and Z_N represent per-channel scale and zero-point parameters. This architecture restricts HBM memory bus traffic strictly to the 4-bit representation, achieving a 75.0% reduction in global VRAM read bandwidth.



Fig. 2. Comprehensive SDSIE hardware telemetry matrix on NVIDIA RTX 5090 Blackwell. **Top-Left:** Energy per token distribution (6.40 J $\rightarrow$ 3.41 J/token). **Top-Right:** Isolated Triton kernel execution latency (77.46 μ s $\rightarrow$ 28.60 μ s). **Bottom-Left:** Llama-3.1-8B SwiGLU MLP layer memory footprint (112.00 MB $\rightarrow$ 31.50 MB). **Bottom-Right:** Speculative decoding throughput acceleration (23.70 $\rightarrow$ 50.52 tok/s).

3.2 Numerically Stable Shannon Entropy Formulation

To monitor generation uncertainty across large vocabulary distributions (e.g., Llama-3's $|\mathcal{V}| = 128, 256$ tokens), naive half-precision softmax calculation triggers floating-point exponent overflow. SDSIE implements a numerically stable float32 log-space formulation:

$$\log p_i = \text{LogSoftmax}(z_i) = z_i - \log \sum_{j \in \mathcal{V}} e^{z_j} \quad (6)$$

The true Shannon entropy $H(X)$ in bits is evaluated as:

$$H(X) = -\frac{1}{\ln 2} \sum_{i \in \mathcal{V}} e^{\log p_i} \cdot \log p_i \quad (7)$$

3.3 Schmitt-Trigger Stochastic Hysteresis Control

To prevent rapid mode-switching (gear hunting) between speculative drafting and single-step decoding on borderline tokens, SDSIE applies an industrial Schmitt-trigger hysteresis filter with an exponential moving average (EMA) smoothing factor $\alpha = 0.35$:

$$\bar{H}_t = \alpha \cdot H(p_t) + (1 - \alpha) \cdot \bar{H}_{t-1} \quad (8)$$

The dynamic draft window k_t is governed by the state transition rule:

$$k_t = \begin{cases} 0, & \text{if } S_{t-1} = \text{Active} \wedge \bar{H}_t > \theta_{\text{high}} \\ 5, & \text{if } S_{t-1} = \text{Fallback} \wedge \bar{H}_t < \theta_{\text{low}} \\ k_{t-1}, & \text{otherwise} \end{cases} \quad (9)$$

where $\theta_{\text{low}} = 0.55$ bits and $\theta_{\text{high}} = 1.25$ bits.

4 EMPIRICAL EVALUATION & HARDWARE TELEMETRY

4.1 Experimental Testbed

All empirical evaluations were executed on a dedicated bare-metal workstation:

- **GPU:** NVIDIA GeForce RTX 5090 32GB (Blackwell Architecture, 512-bit memory bus, 1,792 GB/s bandwidth).

TABLE 1
Layer Latency and Bandwidth on RTX 5090 Blackwell.

Layer / Projection Type	FP16 Base	SDSIE INT4	Reduction
Linear GEMM ($[16 \times 4096]$)	77.46 μ s	28.60 μ s	-63.1%
SwiGLU Block ($[8 \times 4096 \rightarrow 14336]$)	622.4 μ s	155.8 μ s	-74.9%
MLP Weight Bus Traffic / Step	117.44 MB	29.36 MB	-75.0%
Active Core Power	142.1 W	95.3 W	-32.9%

- **CPU / Host:** AMD Ryzen 9 7900X (12 Cores / 24 Threads), 64GB DDR5-6000 RAM.
- **Software Stack:** Ubuntu 24.04 LTS (Linux 6.6 Kernel via WSL2), CUDA 13.3, NVIDIA Driver 610.88, Python 3.12, PyTorch 2.5, OpenAI Triton 3.1.
- **Telemetry Polling:** 100 Hz non-blocking NVML power sampling coupled with high-resolution CPU timestamping (`time.perf_counter`).

4.2 Isolated Kernel Latency & SwiGLU Forward Pass

As detailed in Table 1, the isolated Triton INT4 dequantization GEMM executes in 28.60 μ s compared to 77.46 μ s for unquantized FP16 baseline multiplication under identical tensor dimensions.

When evaluated across a full Llama-3.1-8B SwiGLU MLP block involving three separate linear projections (`gate_proj`, `up_proj`, `down_proj`) and element-wise SiLU activation, SDSIE completes the entire layer forward pass in 155.8 μ s, confirming a 75.0% drop in memory bus data transfer.

4.3 End-to-End Stochastic Generation Case Study

To evaluate the Schmitt-trigger clutch under complex semantic constraints, we benchmarked the engine on a 409-token generation task under strict medieval poetic structure (*Chant Royal*). As visualized in Fig. 1:

- **Deterministic Cadence ($H_{\min} = 0.0001$ bits):** Across continuous rhyming couplets (Steps 90-240), entropy flatlined at 0.0001 bits, holding $k = 5$ speculation continuously with $\approx 100\%$ draft acceptance.

- **Cognitive Rhyme Search ($H_{\max} = 2.98$ bits):** During multi-stanza structural rhyme turns, entropy spiked above 1.25 bits, triggering instant single-step fallback ($k = 0$) and eliminating wasted draft compute.
- **Aggregate Performance:** The run completed 409 tokens in 8.096 s, yielding a sustained throughput of **50.52** tok/s and total energy consumption of 3.41 J/token.

5 CONCLUSION & OPEN-SOURCE AVAILABILITY

SDSIE demonstrates that decoupling precision and speculation windows through software-defined stochastic control resolves the thermodynamic crisis of memory-bound LLM decoding. By fusing sub-byte decompression into on-chip SRAM and governing draft models with an industrial Schmitt-trigger entropy clutch, SDSIE achieves a hardware-verified 46.7% energy reduction on next-generation Blackwell silicon with zero loss in output precision.

All source code, Triton kernels, raw NVML telemetry JSON ledgers, and standalone Web UI harnesses are publicly available under the Apache-2.0 license at <https://github.com/Creepybits/software-defined-stochastic-inference-engine> and permanently archived with CERN/Zenodo (DOI: [10.5281/zenodo.21499379](https://doi.org/10.5281/zenodo.21499379)).

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